2.8.1

EE25BTECH11022 - sankeerthan

Question: If $(-4, 3)$ and $(4, 3)$ are two vertices of an equilateral triangle. Find the coordinates of the third vertex, given that the origin lies in the interior of the triangle.

solution: Let

$$\mathbf{A} = \begin{pmatrix} -4 \\ 3 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 4 \\ 3 \end{pmatrix} and \mathbf{C} = \begin{pmatrix} x \\ y \end{pmatrix} \tag{1}$$

be the vertices of eqilateral triangle.
The direction vector of the segment joining A and B is given by:

$$\mathbf{B} - \mathbf{A} = \begin{pmatrix} 4 - (-4) \\ 3 - 3) \end{pmatrix} = \begin{pmatrix} 8 \\ 0 \end{pmatrix} \tag{2}$$

The side length of equilateral triangle is

$$\|\mathbf{B} - \mathbf{A}\| = \sqrt{(\mathbf{B} - \mathbf{A})^\top (\mathbf{B} - \mathbf{A})} \tag{3}$$

$$= \sqrt{\begin{pmatrix} 8 & 0 \end{pmatrix} \begin{pmatrix} 8 \\ 0 \end{pmatrix}} \tag{4}$$

$$= \sqrt{8(8) + 0(0)} \tag{5}$$

$$= 8 \tag{6}$$

Let **M** be the midpoint of **A** and **B**

$$\mathbf{M} = \frac{\mathbf{A} + \mathbf{B}}{2} \tag{7}$$

$$= \frac{\begin{pmatrix} -4 \\ 3 \end{pmatrix} + \begin{pmatrix} 4 \\ 3 \end{pmatrix}}{2} \tag{8}$$

$$= \frac{\begin{pmatrix} 0 \\ 6 \end{pmatrix}}{2} \tag{9}$$

$$= \begin{pmatrix} 0 \\ 3 \end{pmatrix} \tag{10}$$

Height of equilateral triangle

$$h = \frac{\sqrt{3}}{2} \cdot (8) = 4\sqrt{3} \tag{11}$$

The perpendicular line drawn from **C** meets the line joining the points **A** and **B** at the midpoint $\mathbf{M}$.Thus,the possible values of $\mathbf{C}$ are

$$\mathbf{C} = \begin{pmatrix} 0 \\ 3 + 4\sqrt{3} \end{pmatrix} (or) \begin{pmatrix} 0 \\ 3 - 4\sqrt{3} \end{pmatrix} \tag{12}$$

Since the origin is inside the triangle, choose the lower vertex:

$$\mathbf{C} = \begin{pmatrix} 0 \\ 3 - 4\sqrt{3} \end{pmatrix} \tag{13}$$

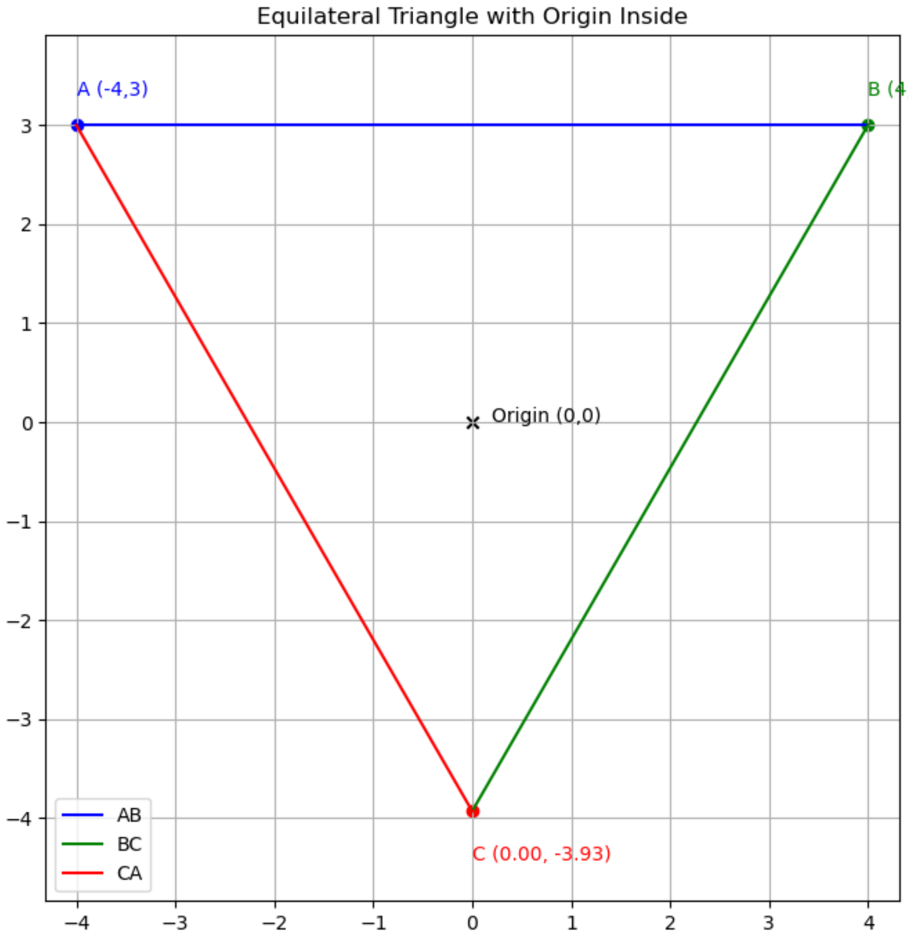

Fig. 0.1